

A rapid and scalable rice blast diagnostic tool - Timely disease identification point-of-care system (TIDIS)

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Africa Rice Center study leader: Geoffrey Onaga
JIC lead partner: Diane Saunders and Pirita Paajanen

Correspondance
Africa Rice, Cote d'Ivoire
G.onaga@cgiar.org

Summary: Rapid and cost-effective sequencing of pathogen genomes has become a fundamental aspect of plant health management in well-resourced settings. This technique needs to be extended to developing countries in order to limit crop losses and improve food security. The combination of pathogen diagnostics and epidemiology with advanced digital disease detection systems has the potential to greatly enhance continuous monitoring, prompt diagnosis, and real-time tracking of crop diseases in regions of the South that have limited access to well-equipped laboratories. AfricaRice and JIC are developing TIDIS, a portable and scalable point-of-care management system for disease identification, under PHI. TIDIS serves as monitoring system that delivers genotypic information of both the host and the pathogen for each field from which the pathogen is isolated and uses this information for epidemiological reconstruction. This facilitates a quick understanding of host-specificity of the pathogen on the crop varieties grown in a specific region, thereby improving the effective management of crop diseases. We conducted a pilot of TIDIS on rice blast, a major yield-limiting pathogen affecting rice worldwide.

Challenge: Rice blast is one of the most widespread and economically important disease of rice worldwide. Early detection, particularly of emerging strains, and timely practical guidance on management are crucial to protect the rice growers. Currently, there is no functioning early warning system for rice blast management in Sub-Saharan Africa (SSA). There is need to establish a comprehensive framework that combines up-to-date field and mobile phone surveillance data, mobile nanopore sequencing diagnostics, and a platform for prompt communication with policy-makers, extension agents, and small-holder farmers. Developing a portable and adaptable disease identification point-of-care management system (TIDIS) will enhance both disease detection and its long-term surveillance for effective management. TIDIS is a pathogen surveillance system designed to rapidly identify the prevalent strains or the diversity within pathogen populations shortly after sample collection. This system is crucial as it addresses the needs of both scientists and agricultural producers by facilitating the prompt identification of pathogen lineages and the susceptible varieties cultivated in a specific area.

Methodology: The study aimed to address the challenges associated with delays in characterizing rice blast pathogen populations by employing direct RNA sequencing of infected leaf samples through a field pathogenomics approach, a method previously applied in Europe to analyze wheat stripe rust pathogen populations from commercial crops. The approach involved collecting rice blast infected leaves from field plots and performing sequencing directly from the samples to determine the genetic lineages to which the samples are associated and susceptible varieties cultivated by farmers for breeding intervention.

Research Results

A total of 84 leaf blast samples were collected from Uganda and Cote d'Ivoire for field pathogenomic analyses. Upon analyzing the samples by aligning the genomes with the global genome collection, we found that the rice blast pathogen populations in Uganda and Cote d'Ivoire have high genetic diversity, with all known lineages present in Uganda and three in Cote d'Ivoire (Figure 1). We also conducted an analysis of the RNA sequence data to identify the different popular varieties cultivated in both countries, from which the infected leaf samples were obtained. The algorithm implemented was tested to ascertain the susceptible varieties by comparing them to 40 sequenced varieties from both nations. At least 10 varieties have been identified as susceptible varieties cultivated by farmers. In addition, we analyzed and refined the effectors by selecting only those exhibiting high polymorphism. Our analysis revealed more than two hundred candidate effectors expressed in planta (Figure 3), including 20 highly polymorphic effectors (Figure 4).

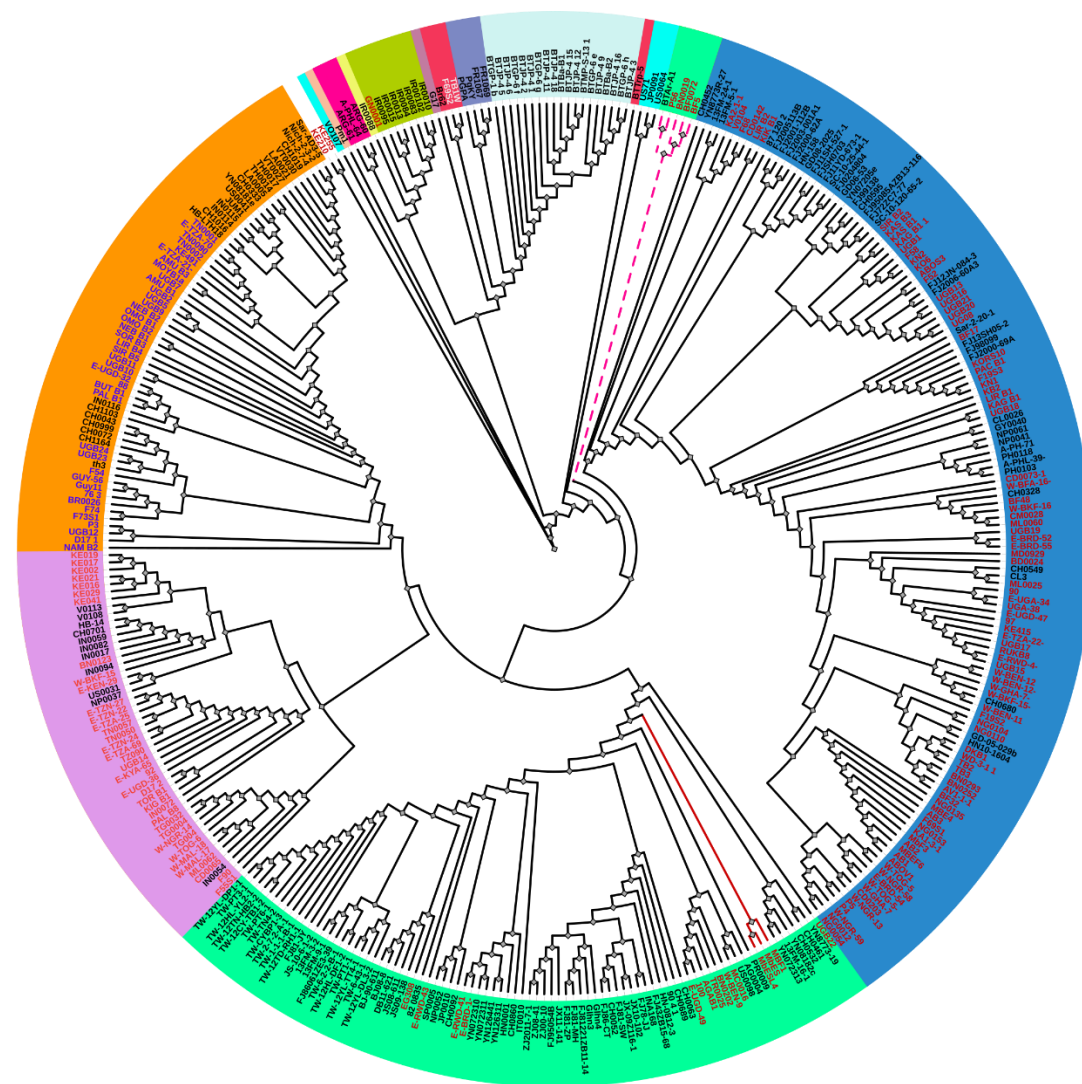


Figure 1

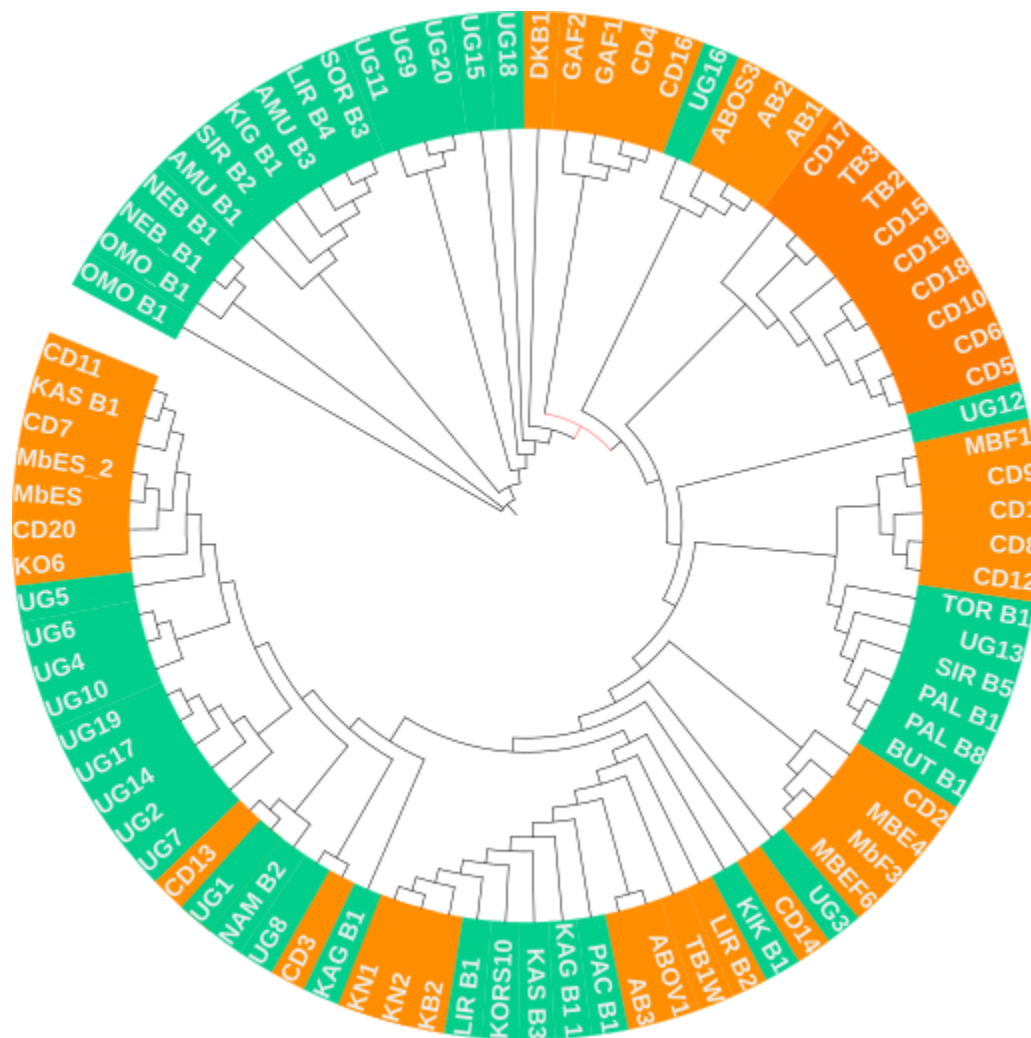


Figure 2

The results of our study hold significance for rice breeders, providing valuable insights into the existing pathogen populations in the two countries and pinpointing susceptible varieties found in farmers' fields. Breeders should consider the potential recall of the susceptible varieties identified in this study for genetic improvement or to replace them with resistant, high-yielding alternatives.

In conclusion, the approach used in this study demonstrates high efficiency relative to conventional methods such as race characterization or genotyping with molecular markers for the analysis of pathogen populations. We plan to continue with customization in 2025, during which MinION sequencing will be adapted for the sequencing of samples in NARs labs. This is expected to facilitate continuous monitoring of pathogen populations and to greatly reduce the time required for the routine monitoring and characterization of crop pathogens.

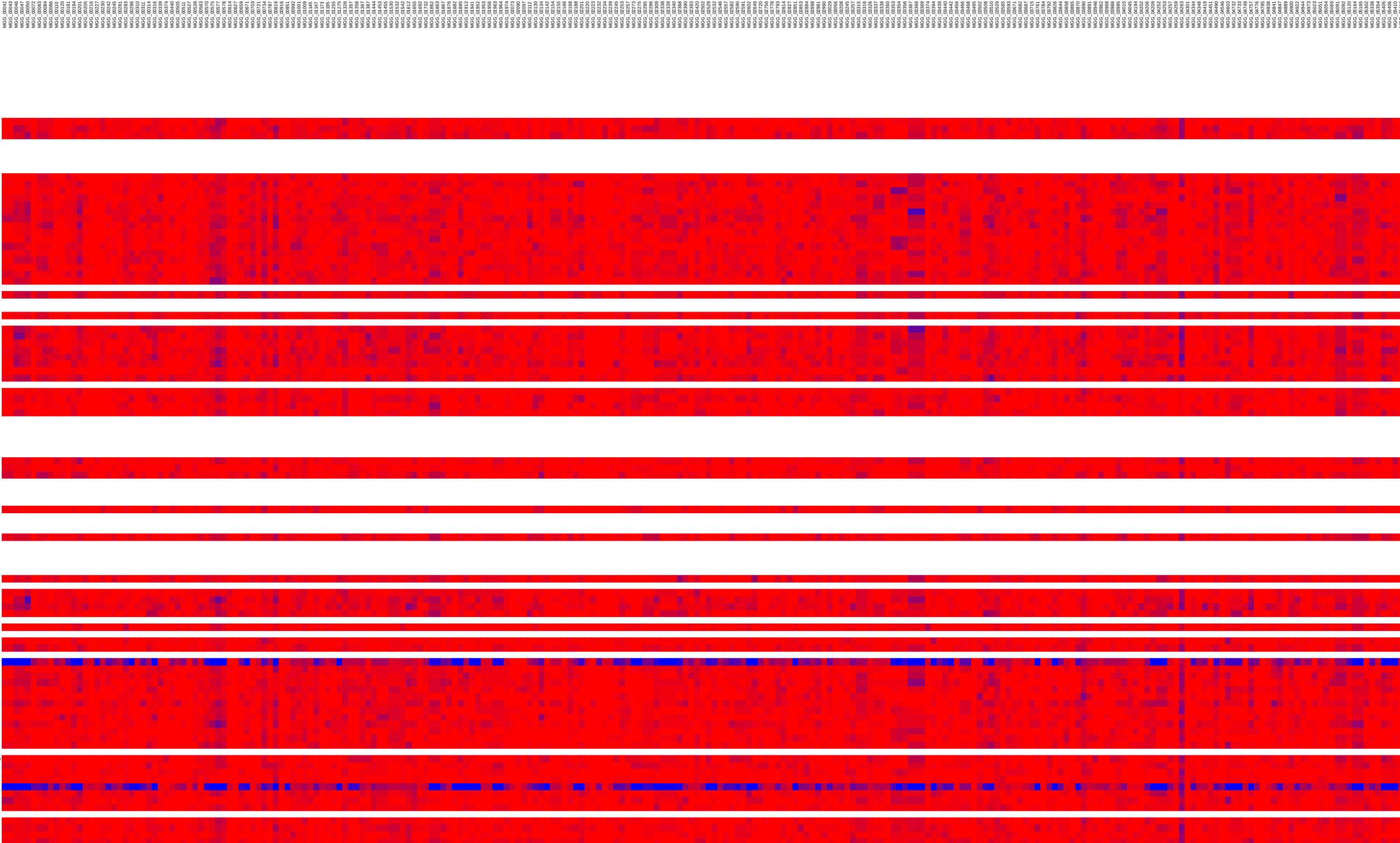
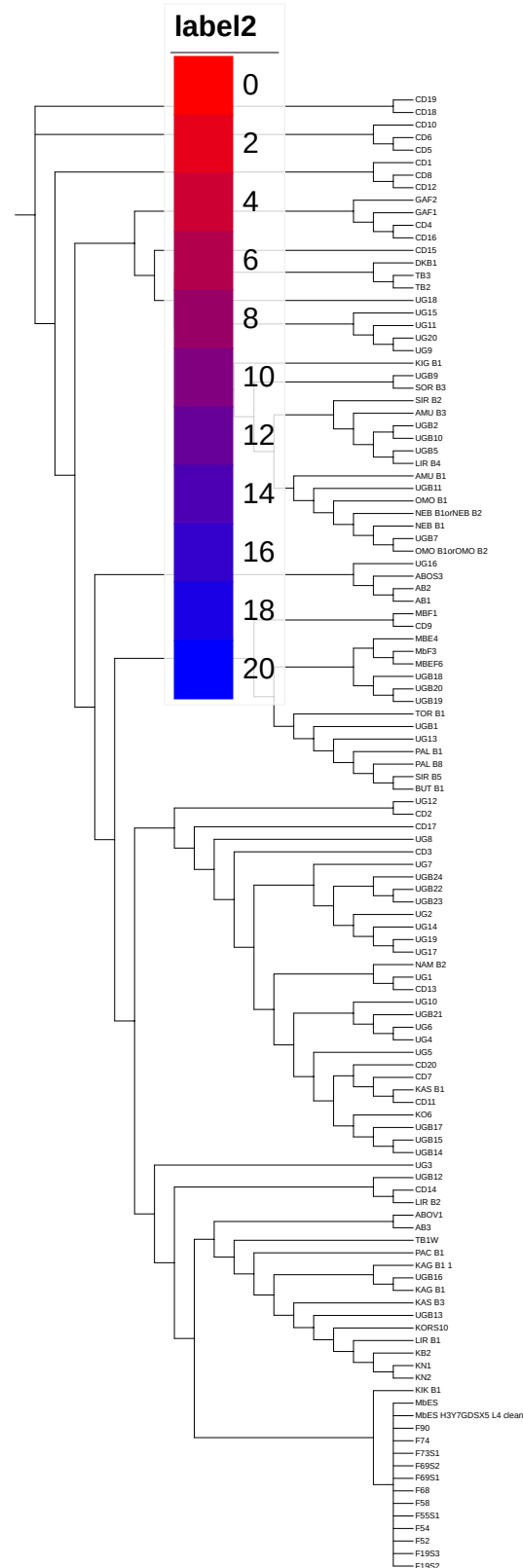
Next steps. The findings of this study are currently being prepared for publication in a peer reviewed journal. Genomic regions/ effectors that are conserved and polymorphic will be further refined for PCR enrichment, followed by test sequencing using the MinION platform. The validation of TIDIS (accuracy and cost-effectiveness) is essential prior to its full-scale deployment. Deployment success will be anchored in training the NARs scientists and NPPOs on its importance and use in 2025/26.

Figure 1 Genome-wide SNP-based NJ tree of 372 *P. oryzae* and *M. grisea* isolates (smaller clusters). *P. oryzae* isolates originating from Africa are in red, purple, and white fonts. The lineages are distinguished by color, ranging from Orange to Blue. Dotted lines indicate a recombining group identified in previous studies (Barragan et al. 2024).

Figure 2 Genome-wide SNP-based NJ tree of popular varieties aligned with pathogen associated RNA sequence data. The varieties are distinguished by color, ranging from Orange (Cote d'Ivoire) to green (Uganda)

Figure 3 (pdf): Analysis of *P. oryzae* candidate effectors. The tree and heatmap show the relationship and prevalence of candidate effectors among strains collected from different locations in Uganda and Cote d'Ivoire. A stringent E value of $1e-50$ was employed to identify potential homologs, and only the top BLAST hit was incorporated into the heatmap. The color spectrum indicates the dominance of a specific effector.

Figure 4 (pdf): Analysis of *P. oryzae* candidate effectors. The tree and heatmap show the relationship and prevalence of highly polymorphic candidate effector among strains collected from various location in Uganda and Cote d'Ivoire.



Tree scale: 1

